Zhiming Huang

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EMPLOYMENT

Postdoc at Paul G. Allen School of Computer Science & Engineering — Dec 2025 - Now

- **University of Washington**, WA, USA
- Advisor: Prof. Jamie Morgenstern

EDUCATION

University of Victoria, BC, Canada — Sep 2018 - Aug 2025

- **Ph.D.** in Computer Science (Aug 2020 - Aug 2025)
- Advisor: Prof. Jianping Pan
- Dissertation: Swap-Regret-Minimizing Bandits for Distributed Network Optimization
- **M.Sc.** in Computer Science (Sep 2018 - Aug 2020)
- Advisor: Prof. Jianping Pan
- Thesis: Thompson sampling-based online decision making in network routing

B.Eng. in Communication Engineering — Sep 2014 - Jun 2018

- **Northwestern Polytechnical University**, Shaanxi, China

RESEARCH INTERESTS

Online and game-theoretic learning for sequential decision-making under limited feedback, with applications to networking and distributed systems.

PUBLICATIONS

(α-β) denotes alphabetical author order.

CONFERENCE PUBLICATIONS

[1] **Z. Huang**, J. Morgenstern, A. Roth and C. J. Zhang (α-β), Instance-Adaptive Online Multicalibration, *Advances in Neural Information Processing Systems (NeurIPS 2026)*, accepted.

[2] **Z. Huang**, B. Hu and J. Pan, Bridging the Regret Gap in Combinatorial Thompson Sampling: Worst-Case Guarantees and Algorithmic Refinement, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2026.

[3] Q. Zhang, **Z. Huang**, J. Zhao, A. Ahangarpour and J. Pan, Faster Exploration and Exploitation for Communication Environment Awareness in Starlink, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2026.

[4] **Z. Huang** and J. Pan, Faster Convergence for Unknown-game Bandits, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2025.

[5] **Z. Huang** and J. Pan, Adversarial Semi-Bandits with Moving Arms, *IEEE International Conference on Computer Communications (INFOCOM)*, May 2025.

[6] B. Hu, **Z. Huang**, T. Zhang, M. Lécuyer and N. Hegde, Connecting Thompson Sampling and UCB: Towards More Efficient Trade-offs Between Privacy and Regret, *International Conference on Machine Learning (ICML)*, May 2025.

[7] Q. Zhang, **Z. Huang**, J. Zhao, and J. Pan, A Congestion Control Test Suite for Real-Time Communication, *ACM Multimedia Systems Conference (MMSys)*, Apr 2025.

[8] **Z. Huang**, B. Hu, and J. Pan, Gaussian Randomized Exploration for Semi-bandits with Sleeping Arms, *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, Dec 2024.

[9] F. Amri, **Z. Huang**, K. Liu, and J. Pan, Energy-aware Inter-data Center VM Migration over Elastic Optical Networks, *IEEE International Conference on Communication (Globecom)*, Dec 2023.

[10] **Z. Huang** and J. Pan, A Near-optimal High-probability Swap-Regret Upper Bound for Multi-agent Bandits in Unknown General-sum Games, *Uncertainty of Artificial Intelligence (UAI)*, Aug 2023.

[11] **Z. Huang**, K. Liu and J. Pan, End-to-end Congestion Control as Learning for Unknown Games, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2023.

[12] **Z. Huang**, B. Hu, and J. Pan, Poster: Multi-agent Combinatorial Bandits with Moving Arms, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2021.

[13] B. Hu, Y. Chen, **Z. Huang**, N. Mehta, and J. Pan, Intelligent Caching Algorithms in Heterogeneous Wireless Networks with Uncertainty, *IEEE International Conference on Distributed Computing Systems (ICDCS)*, Jul 2019.

[14] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: An Enhanced QR Code-based Secure E-coupon Transaction Framework, *IEEE International Conference on Communication (ICC)*, May 2019.

PUBLICATIONS

PREPRINTS

[1] **Z. Huang**, J. Morgenstern, A. Roth and C. J. Zhang (α-β), Instance-Adaptive Online Multiclass Multicalibration, *arXiv:2610.15954*, Oct 2026. Preprint.

JOURNAL PUBLICATIONS

[1] W. Yang, L. Cai, S. Shu, A. Sepahi, **Z. Huang** and J. Pan, QoS-driven Contextual MAB for MPQUIC Supporting Video Streaming in Mobile Networks, *IEEE Transactions on Mobile Computing (TMC)*, Nov 2024.

[2] **Z. Huang** and J. Pan, Game-theoretic Bandits for Network Optimization with High-probability Swap-regret Upper Bounds, *IEEE/ACM Transactions on Networking (TON)*, Jul 2024.

[3] B. Hu, **Z. Huang**, N. Mehta, and N. Hegde, Near-Optimal Algorithms for Differentially Private Online Learning in a Stochastic Environment, *arXiv*, Jul 2024.

[4] K. Liu, J. Wang, **Z. Huang**, and J. Pan, Sampling-based Job Placement and Load Balancing for Heterogeneous Deep Learning Clusters, *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, Apr 2024.

[5] **Z. Huang** and J. Pan, Distributed Learning of Unknown Games for HetNet Selection, *IEEE Transactions on Network Science and Engineering (TNSE)*, Jan 2024.

[6] **Z. Huang**, Y. Xu, and J. Pan, TSOR: Thompson Sampling-based Opportunistic Routing, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 11, pp. 7272-7285, May 2021.

[7] **Z. Huang**, B. Hu, and J. Pan, Caching by User Preference with Delayed Feedback for Heterogeneous Cellular Networks, *IEEE Transactions on Wireless Communications (TWC)*, vol. 20, no. 3, pp. 1655-1667, Mar 2021.

[8] R. Liu, J. Song, **Z. Huang**, and J. Pan, EQRC: A Secure QR Code-based E-coupon Framework Supporting Online and Offline Transactions, *Journal of Computer Security (JCS)*, vol. 28, no. 5, pp. 577-605, Sep 2020.

[9] **Z. Huang**, R. Zhang, J. Pan, Y. Jiang, and D. Zhai, A Framework of Multipath Clustering based on Space Transformed Fuzzy c-Means and Data Fusion for Radio Channel Modeling, *IEEE Transactions on Vehicular Technology (TVT)*, vol. 69, no. 1, pp. 4-15, Jan 2019.

TEACHING EXPERIENCE

INSTRUCTOR

CSC111: Fundamentals of Programming with Engineering Applications Spring 2024

TEACHING ASSISTANT

- **Lab and Tutorial Instructor for CSC361: Computer Communication and Networks**
- **Tutorial Instructor for CSC360: Introduction to Operating Systems**

PROFESSIONAL ACTIVITIES

PROGRAM COMMITTEE

- Shadow TPC member for ACM CoNEXT 2025

REVIEWER

- IEEE Transactions on Wireless Communications
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Signal Processing

AWARDS & SCHOLARSHIPS

Canada NSERC Postdoctoral Fellowship 2025
IEEE INFOCOM 2025 Travel Award 2025
British Columbia Graduate Scholarship 2024
UVic President's Fellowship in Research-enriched Teaching 2023
ICDCS 2023 Travel Award 2023
UAI 2023 Scholarships 2023
Melva J. Hanson Graduate Scholarship 2022
IEEE ICDCS 2021 Best Poster Award 2021
Howard E. Petch Research Scholarship 2020